

## Step-by-Step Guide for Creating and Configuring a DynamoDB Table.

We will be creating an Amazon DynamoDB Table to support our project data. The steps to create a DynamoDB Table are outlined below, each accompanied by a brief explanation to clarify its importance and relevance to our use case.

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### Step 1 – Sign in and Navigate to DynamoDB.

- **Sign in** to the **AWS Management Console**, locate **“Services”** in the top menu (left side), **search** and **click “DynamoDB”** as shown in figure 1.

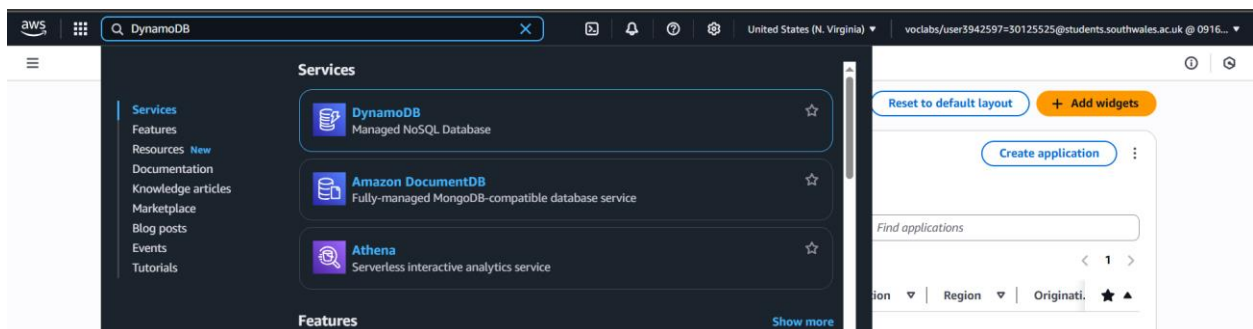


Figure 1. Navigating to DynamoDB.

### Step 2 – Importing a DynamoDB Table from S3 Bucket.

- Click **“Imports from S3”** on the left panel as marked in figure 2.

- If we want to import any data from the S3 bucket we must use the easy feature mentioned since it is easy to navigate.

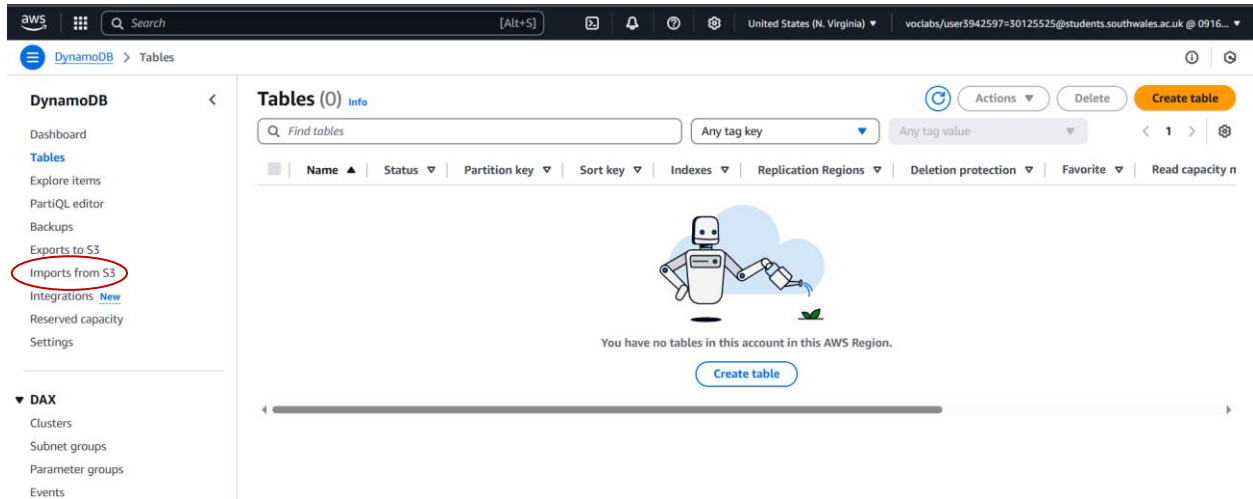


Figure 2. Importing data from S3 bucket.

- Now, click “**Import from S3**” as shown in the figure 3.

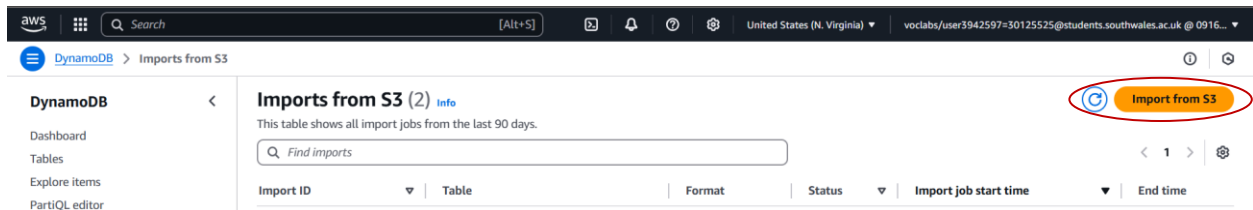


Figure 3. First step to import JSON file to a DynamoDB Table.

### Step 3 – Copy S3 URI path from the uploaded JSON file in S3 bucket.

- Open S3 Dashboard in a new tab, inside the newly created bucket click the JSON file.
- Click “**Copy S3 URI**” from the desired S3 bucket as shown in figure 4 below.
- By copying the URI path, it will make sure that we are importing the specific and desired file.

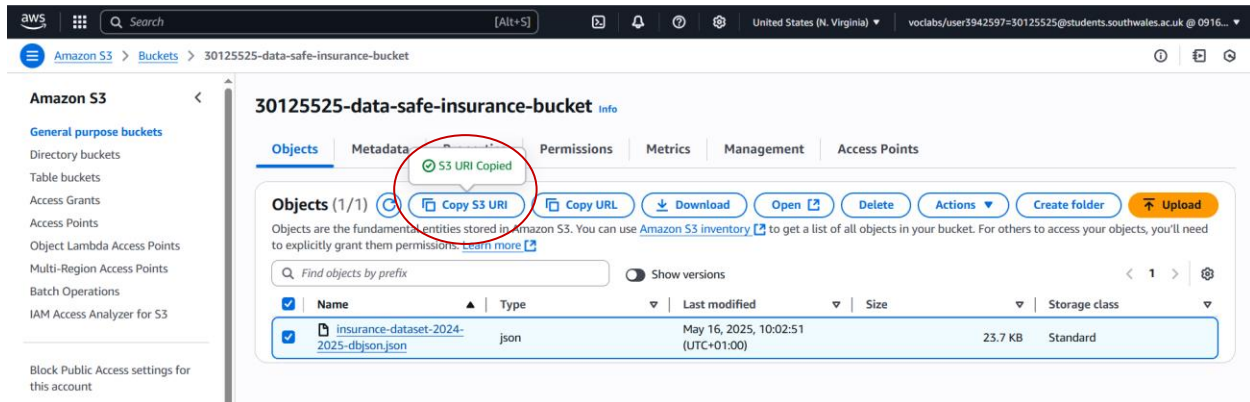


Figure 4. Copying S3 URI path.

## Step 4 – Pasting the URI path in DynamoDB.

- Next, we must **paste** the copied “S3 URI” in the Source S3 URL as shown in figure 5 and click next.

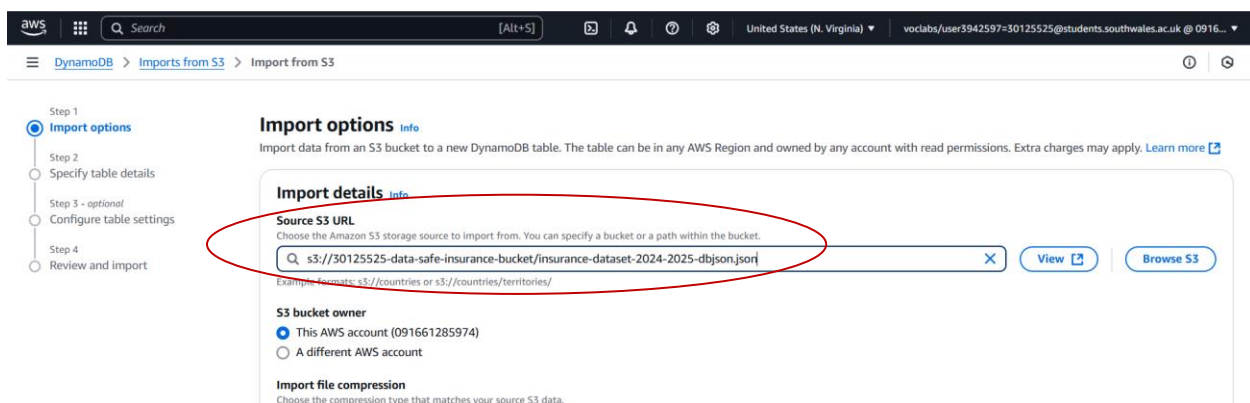


Figure 5. Pasting the URI path in DynamoDB source URL.

## Step 5 – Inputting table details and keys.

- Now, enter the desired table name you want. Here, we are giving our **table name** as “30125525-data-safe-insurance-db” as shown in Figure 6.
- Next, we must set Primary Keys and Sort Keys. Here, for our data **Primary key** is “Client Username” and “Sort Key” is “Client Country” of data type “String.”
- The partition key uniquely identifies each item and determines data distribution across storage nodes.
- Click **Next**.

**Specify table details**

**Table details** [Info](#)  
DynamoDB is a **schemaless** database that requires only a table name and a primary key when you create the table.

**Table name**  
This will be used to identify your table.  
  
Between 3 and 255 characters, containing only letters, numbers, underscores (\_), hyphens (-), and periods (.).

**Partition key**  
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.  
  1 to 255 characters and case sensitive.

**Sort key - optional**  
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.  
  1 to 255 characters and case sensitive.

[Cancel](#) [Previous](#) [Next](#)

Figure 6. Specifying table detail.

## Step 6 – Selecting Table Settings.

- Here, we will go ahead without customizing the settings as shown in figure 7.
- Click **Next**.
- In default settings -
  - **Capacity mode** is “On-demand” as it automatically adapts to traffic volumes.
  - **Deletion Protection** is off initially. We will enable it after creating our table.
  - **Encryption Key Management** is owned by AWS and is not charged.
  - We do not require any GSI now. It allows to query data using attributes other than the primary key.

**Configure table settings - optional**

**Table settings**

☒ **Default settings**  
The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose "Customize settings".

☐ **Customize settings**  
Use these advanced features to make DynamoDB work better for your needs.

**Default table settings**  
These are the default settings for your new table. You can change some of these settings after creating the table.

| Setting                      | Value                    | Editable after creation |
|------------------------------|--------------------------|-------------------------|
| Table class                  | DynamoDB Standard        | Yes                     |
| Capacity mode                | On-demand                | Yes                     |
| Maximum read capacity units  | -                        | Yes                     |
| Maximum write capacity units | -                        | Yes                     |
| Local secondary indexes      | -                        | No                      |
| Global secondary indexes     | -                        | Yes                     |
| Encryption key management    | Owned by Amazon DynamoDB | Yes                     |

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Figure 7. Selecting default settings.

## Step 7 – Review and import.

- Finally, we must review our settings and then click **“Import”** as depicted in figure 8.

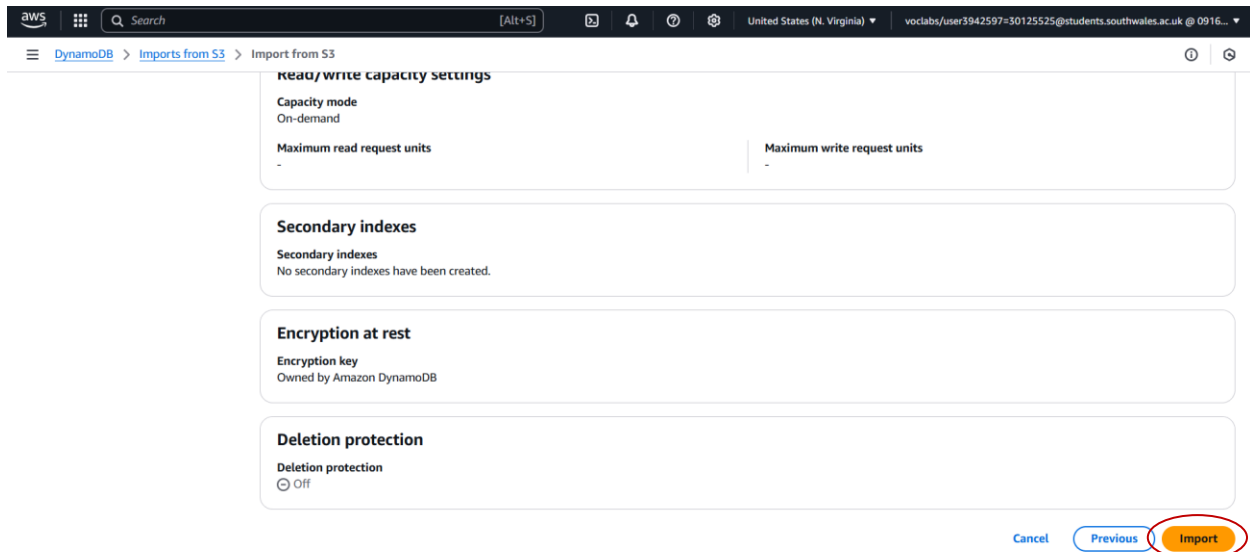


Figure 8. Final step to import DynamoDB Table from S3.

- After importing the data, we can see the table **“30125525-data-safe-insurance-db”** in the tables section as shown in the figure 9.

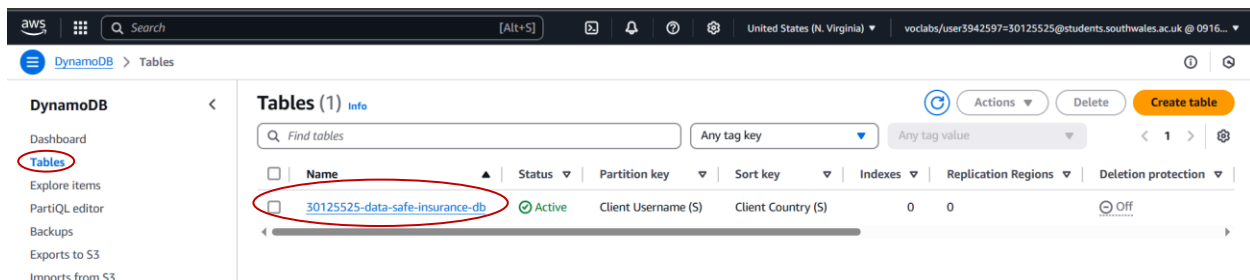


Figure 9. Snippet of DynamoDB Table.

## Step 8 – Enable Deletion Protection.

- Click the newly created table and under settings go to **“Deletion protection”** then click **“Turn on”** as depicted in figure 10.
- It provides an extra layer of security for critical resources, preventing accidental deletion of table.

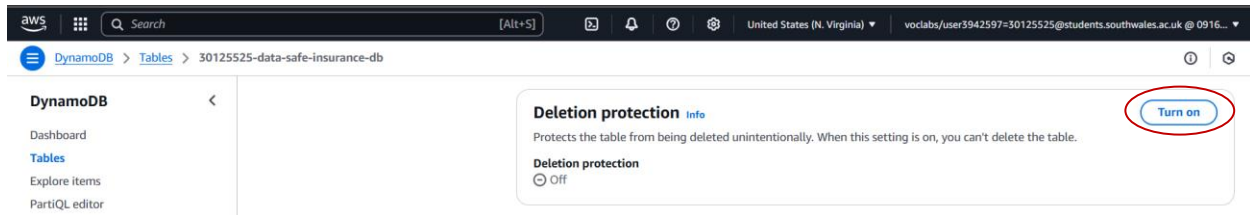


Figure 10. Enabling Deletion Protection

## Step 9 – Turn on Point-in-time-recovery (PITR).

- Under “**Backups,**” click “**Edit**” for PITR as shown below in figure 11.

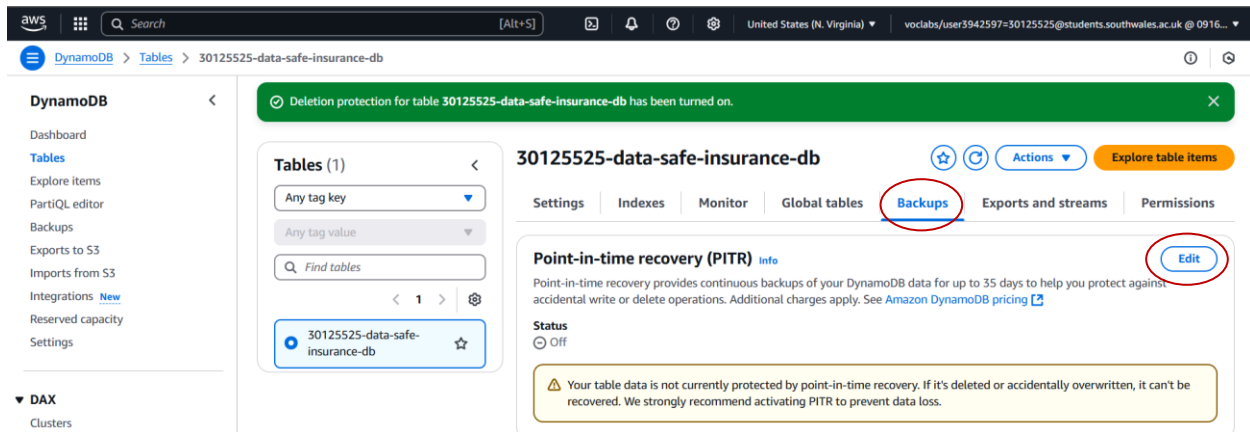


Figure 11. Enabling PITR.

- Click the “**Turn on point-in-time recovery**” to enable it and put 30 days as shown in figure 12 and then click “**Save Changes**”
- Point-in-time recovery provides continuous backups of your DynamoDB data for up to 35 days to help you protect against accidental writing or delete operations.

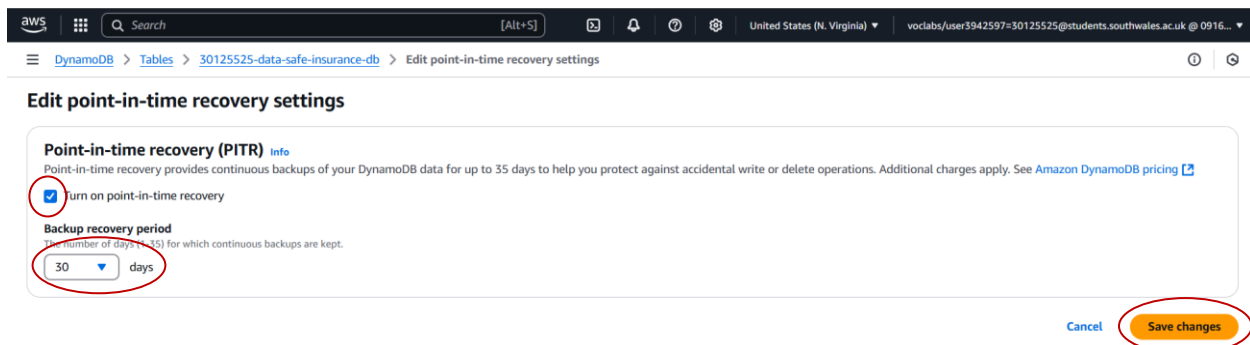


Figure 12. Saving the changes.

## Step 10 – Checking the contents of our Table.

- Finally, after all the steps have been completed click “**Explore table Items**” to open our table.

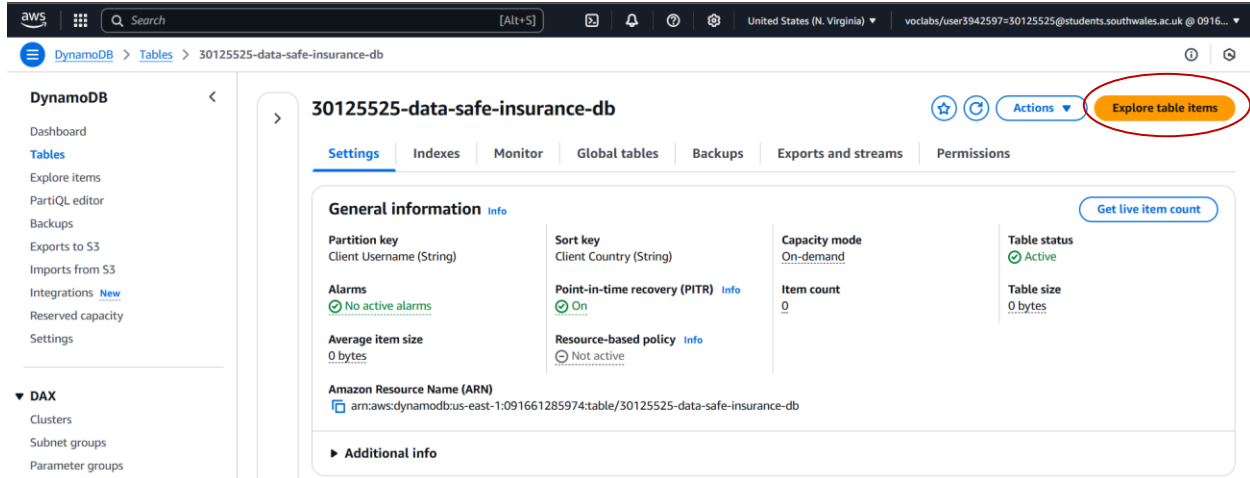


Figure 13.

- We can see our JSON data loaded as a DynamoDB table as shown in figure 14.
- We have 40 rows and 11 columns on our data table.

The screenshot shows the 'Explore Items' view of the '30125525-data-safe-insurance-db' table. It displays 40 items returned. The table has the following columns: Client Username (String), Client Country (String), Annual Premium 2024, Annual Premium 2025, Claim Amount 2024, Claim Amount 2025, and Claim. The data is sorted by Client Username.

| Client Username (String) | Client Country (String) | Annual Premium 2024 | Annual Premium 2025 | Claim Amount 2024 | Claim Amount 2025 | Claim |
|--------------------------|-------------------------|---------------------|---------------------|-------------------|-------------------|-------|
| Rodriguez_M              | US                      | 0                   | 1200                | 0                 | 0                 | 0     |
| Mueller_F                | DE                      | 640                 | 680                 | 2800              | 2880              | 1     |
| Bernard_T                | FR                      | 0                   | 480                 | 0                 | 2300              | 0     |
| Mountain_Tours           | DE                      | 1800                | 1950                | 64000             | 64100             | 1     |
| Mountain_Tours           | US                      | 380                 | 420                 | 0                 | 1228              | 0     |
| Ahmed_Family             | UK                      | 720                 | 778                 | 0                 | 4300              | 0     |
| Schmidt_GmbH             | DE                      | 1400                | 1512                | 18000             | 30600             | 1     |
| Martinez_R               | US                      | 580                 | 625                 | 0                 | 2100              | 0     |
| Thompson_K               | US                      | 490                 | 543                 | 1200              | 2280              | 1     |
| Moreau_Family            | FR                      | 1300                | 1411                | 35000             | 35153             | 1     |
| Waterston_Group          | UK                      | 2400                | 2550                | 110000            | 102335            | 2     |
| Oxford Properties        | UK                      | 3600                | 3870                | 120000            | 246284            | 1     |

Figure 14. Final DynamoDB Table.

## References –

- [1] Amazon Web Services, “Step 1: Create a table in DynamoDB,” Amazon DynamoDB Documentation, 2023. [Online]. Available: <https://docs.aws.amazon.com/dynamodb/latest/developerguide/GettingStarted.CreateTable.html>. [Accessed: 23 May 2025].
- [2] Amazon Web Services, “Exporting DynamoDB table data to Amazon S3,” Amazon DynamoDB Documentation, 2024. [Online]. Available: <https://docs.aws.amazon.com/dynamodb/latest/developerguide/S3DataExport.html>. [Accessed: 23 May 2025].
- [3] Amazon Web Services, “Getting started with DynamoDB,” Amazon DynamoDB Documentation, 2024. [Online]. Available: <https://docs.aws.amazon.com/dynamodb/latest/developerguide/GettingStarted.html>. [Accessed: 23 May 2025].